

Evaluation methods

Setup:

1. Prepare two datasets:
 - (a) Choose one real dataset corresponding to a binary classification problem.
 - (b) Create an artificial dataset in the following way:

$$y_i \sim \text{Bern}(p_i),$$

where

$$p_i = \frac{1}{1 + \exp[-(\alpha + \beta^T x_i)]},$$

for $i = 1, \dots, n$, $x_i \sim N(0, I)$ and $\beta = (b, b, b, b, b, 0, \dots, 0)$ (there are 5 relevant variables and k irrelevant variables).

The parameters n, b, k , and α control, respectively, the sample size, signal strength, number of irrelevant variables, and class imbalance. We will vary them in simulations.

2. How will increasing α influence the imbalance in the dataset?

For a given α , what is the expected class proportion $\mathbb{P}(Y = 1)$? How does it determine the class imbalance?

3. Consider two classification models: logistic regression and classification tree. Which model do you expect to be more stable under repeated sampling?

Tasks:

1. For one chosen setting of parameters (e.g. $n = 1000$, $b = 1$, $k = 20$, $\alpha = 1$), estimate the misclassification error (using threshold 0.5) of both models using the following evaluation schemes:
 - (a) Refitting, i.e. use the whole dataset for both training and testing.
 - (b) 10-fold cross-validation.
 - (c) Bootstrap method.
 - (d) Bootstrap 0.632.

Compare the obtained estimates. Which method gives the most optimistic estimate? Which one seems the most reliable? Explain the differences.

2. Split data into training (50%) and validation (50%) sets. Draw ROC curve and precision-recall curve using the validation data for one of the models. Try different values of parameters for artificial dataset $n \in \{100, 1000\}$, $b \in \{0.5, 1\}$, $k \in \{5, 50\}$ and class imbalance $\alpha \in \{0, 2\}$. For which α value is the precision-recall curve more informative than the ROC curve?

3. Split data into training (50%) and validation (50%) sets. Fit the logistic regression model using training and calculate accuracy and balanced accuracy on validation data using classification rule

$$\mathbb{P}(y_i = 1|x_i) > t,$$

where t is a threshold. Draw plots showing how accuracy and balanced accuracy depend on $t \in [0, 1]$. On the accuracy plot, draw a line at $t = 0.5$; while for the balanced accuracy, draw a line at $t = \frac{1}{n_{\text{train}}} \sum_{i=1}^{n_{\text{train}}} y_i$. Why the accuracy / balanced accuracy tends to have a maximum around those points?

How would the optimal threshold change if false positives and false negatives had different costs?